REPORT

INTRODUCTION

ONLINE RETAIL AND E-COMMERCE SALES ANALYSIS

This project aims to analyze transactional data from an online retail store to uncover patterns in sales, customer behavior and revenue distribution. Using data analysis and visualization techniques, which enables one to generate insights that can support the business decision making and improve performance.

Proposed by: Mwaura Ngugi-677652

Course: Data Science and Analytics

Date: 3/6/2026

Submitted to: Prof Austin Odera

DATASET DESCRIPTION

Dataset Source: Kaggle

Dataset name: Online Retail II UCI

Number of rows and columns: (1017177, 38)

URL: https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci/data

GitHub link- https://github.com/mwaurangugi57-maker/DSA1080A-SS2026-Mwaura-Ngugi-ONLINE-RETAIL-AND-E-COMMERCE-SALES-ANALYSIS

DATA ANALYSIS:

RESEARCH QUESTIONS :

What are the revenue trends? The monthly revenue data shows fluctuating but generally increasing sales overtime with clear patterns of growth and decline across different periods

Key Observations:

- **Initial growth phase (2010**)-Revenue increases from 548000(February) to 777000(March 2010) which shows early growth and rising customer activity.
- **Mid-Year Fluctuations**-Revenue remains relatively unstable between April-August, Values fluctuate between 650000 and 710000 which indicates moderate but inconsistent demand.
- **Strong Growth Towards Year End**-Significant increase from 886000(Sep 2010) to 1.45million (Nov 2010)
- **Seasonal Peak**-Revenue peaks around November and followed by a drop in December which suggests high activity before holidays.

Which are the top selling products? The product "WORLD WAR 2 GLIDERS ASSTD DESIGNS" is the highest selling item with over 106000 units sold, significantly outperforming all other products. It is followed by "WHITE HANGING HEART T-LIGHT HOLDER" and "ASSORTED COLOR BIRD ORNAMENT" which also shows strong sales performance.

Key Observations:

- A few products consistently rank at the top, indicating high customer demand for specific items.
- Sales gradually decline across lower-ranked products, showing most items contribute less to total sales.

- Many of the top performing items are decorative suggesting a common purchasing theme.

How do the customers behave when purchasing various products? The results show that customer spending is **highly concentrated**, with a small number of customers contributing a large share of total sales. The top customer dominates both in quantity purchased and total spending, suggesting bulk purchasing behavior or potential business-to-business transactions.

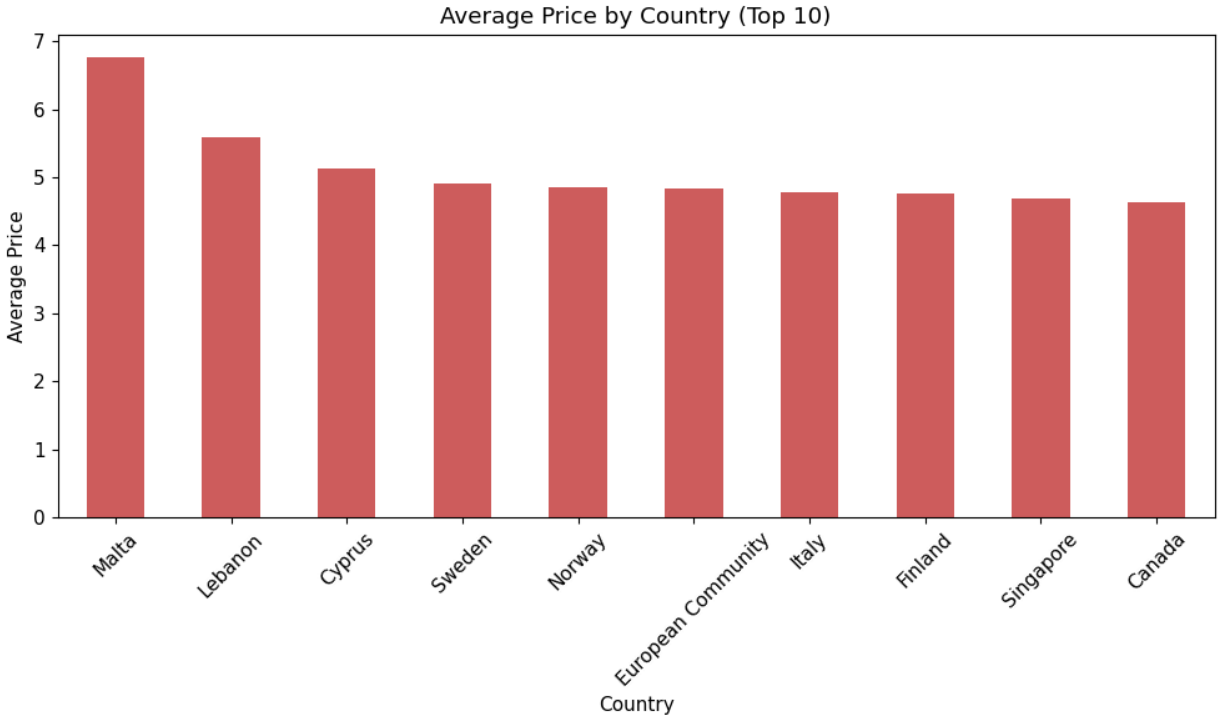
This shows:

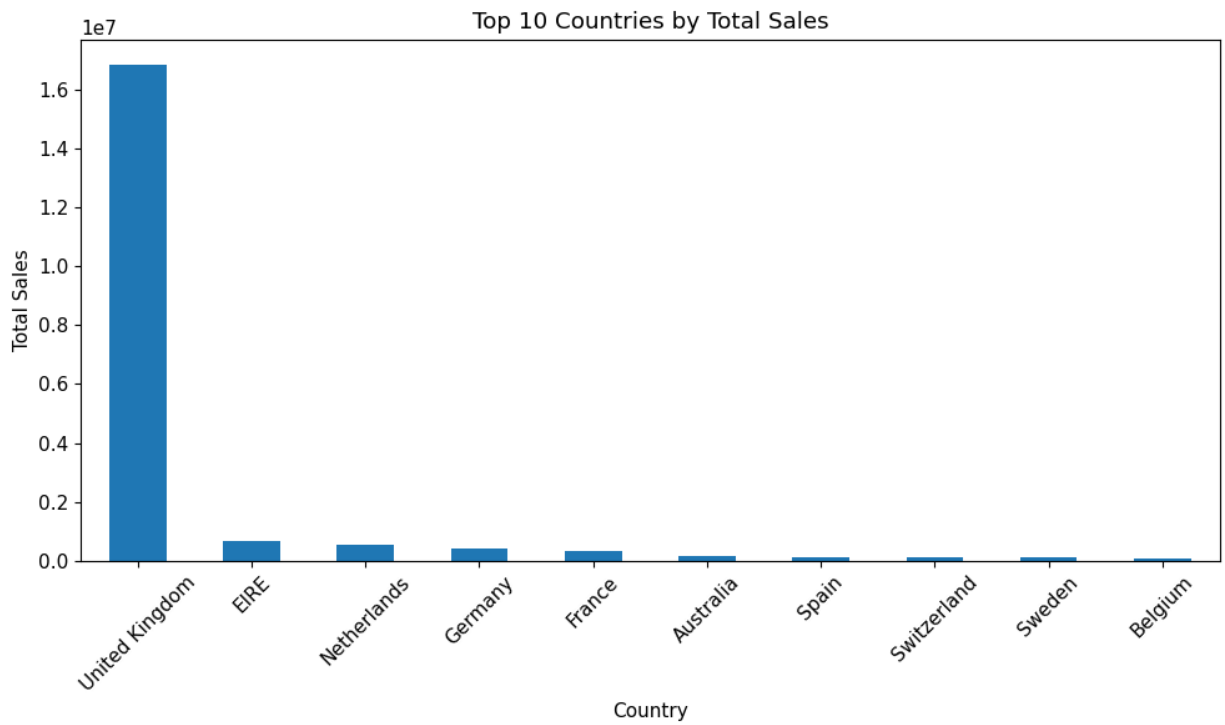
- The business depends heavily on a few high value customers.
- Most customers contribute relatively small amounts individually
- Customer behavior varies significantly across the dataset

How do sales compare in respect to countries? The results indicates that sales are not evenly distributed geographically, with a few countries dominating the total revenue. This suggests that the business has stronger market penetration and customer engagement in specific regions particularly in parts of Europe.

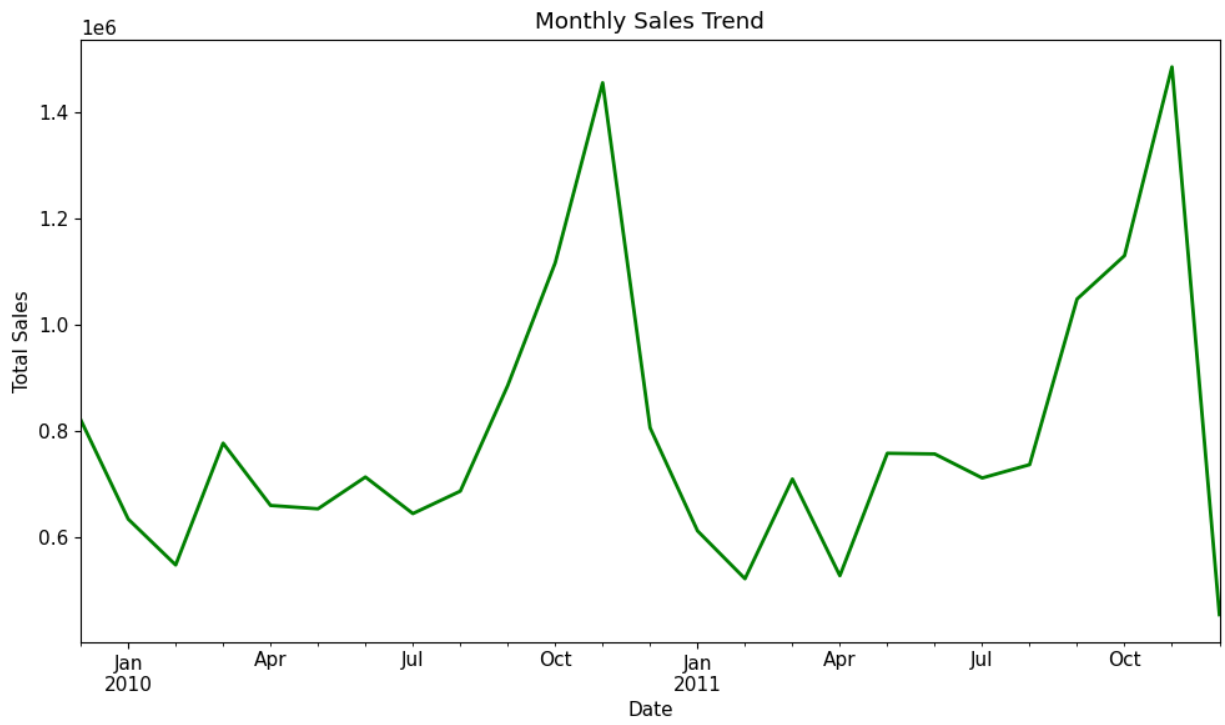
VISUALIZATIONS

Some countries have higher average prices, which may indicate premium products or pricing difference

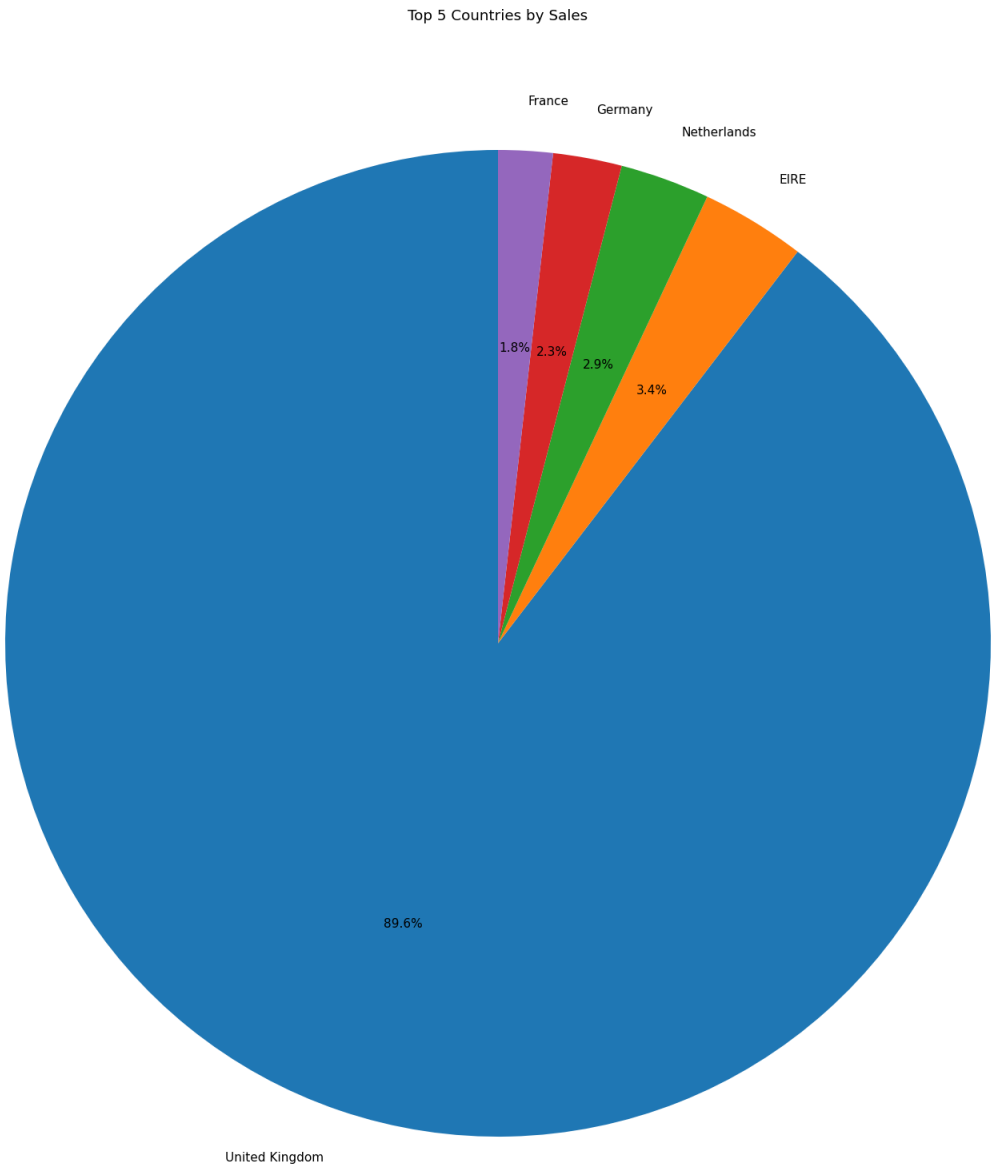




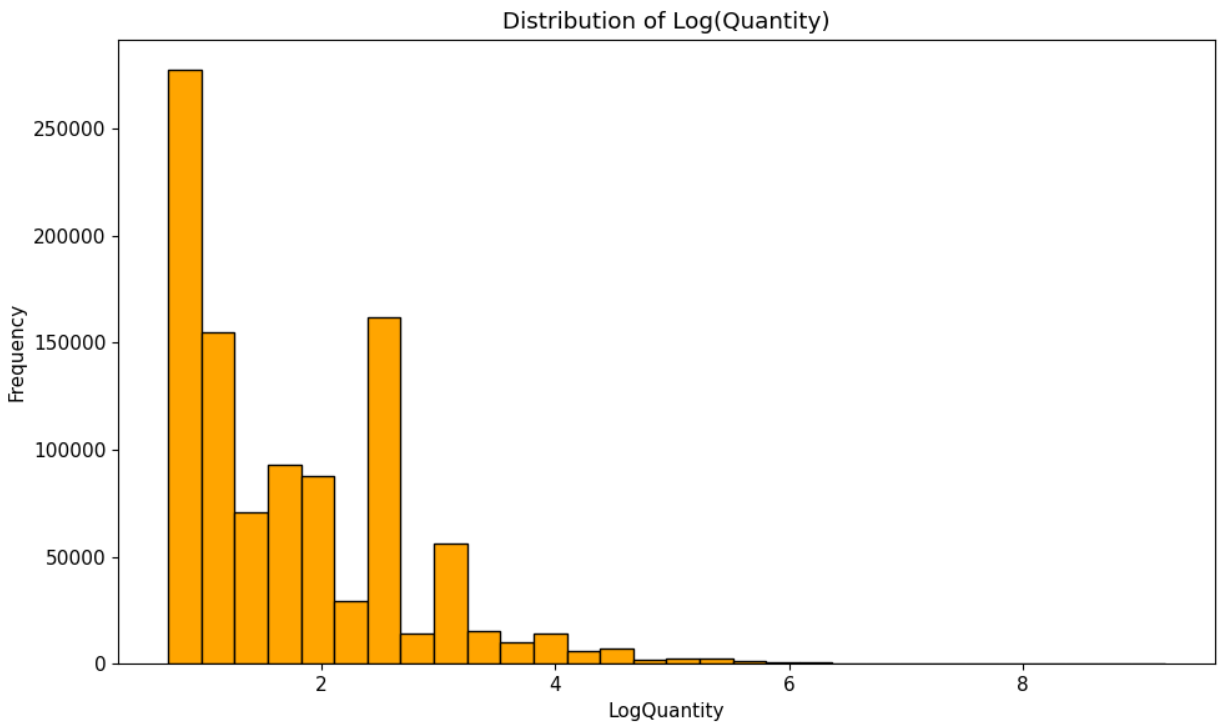
The quantity purchased is highest in The United Kingdom this shows that there is high demand in The United Kingdom as compared to other countries



The line plot shows that most of the quantity purchased was done in December since there is peak in sales around December which suggests that demand is seasonal.



The pie chart shows the propotion of total Sales contributed by each country. It is evident that a few countries dominate Sales with the top contributers being the UK which accounts the largest share which suggests that the #business relies heavily on specific markets.



The original quantity was highly skewed with many outliers so applying a logarithmic transformation compresses the scale and reveals underlying patterns more clearly, making the data more suitable for analysis.

Therefore, this indicates that most purchases cluster around moderate values, indicating that customers typically purchase in small to medium quantities, while large bulk purchases are less.

KEY INSIGHTS

1. A few products consistently rank at the top, indicating high customer demand for specific items

2. The business heavily depends on a few high valued customers

3. The business receives most of its revenue from the UK indicating the business dominates the UK market more than the rest of the regions which indicates sales are not evenly distributed

4. Revenue peaks around November and followed by a drop in December which suggests high activity before holidays.

5. The average transaction value is higher on weekdays than on the weekends this shows that customers tend to purchase more on the weekdays than on the weekends since small quantities are purchased on the weekends

RECOMMENDATIONS:

- The business should strengthen its presence in high performing markets such as the UK while also exploring strategies to expand in underperforming regions to diversify revenue sources to reduce risk.
- The presence of extremely high-value customers suggests an opportunity to focus on customer retention and relationship management. These customers are critical to revenue generation and should be prioritized through targeted strategies.
- The consistent increase in revenue toward the end of each year highlights the importance of seasonal demand in driving sales. The business appears to rely heavily on peak periods, particularly around November, for revenue growth.

CONCLUSION:

Overall, the findings suggest that the business relies heavily on high performing customers, products, and markets. leveraging these insights can help improve decision making in areas such as marketing, inventory management and customer engagement.